M&AE 4070/5070 - Dynamics of Flight Vehicles

Chapter 4 - Take-off and Landing

Companion Notes for Case Study 4

1 Introduction

Take-off and landing are characterized by safety and performance issues:

- **Safety:** The aircraft is very close to the ground and at low velocity. This translates into a much higher risk of having an accident than the other flight stages.

Flight Stage	Accident rate τ	Time spent in stage T	Ratio τ/T
Take off	29%	2%	14.5
Landing	41%	4%	10.3
Total (take-off/landing)	70%	6%	11.66
Cruise	5%	60%	0.083

It comes without surprise that take-off and landing will be subjected to a very high number of regulations, with numerous constraints, in particular on velocity, that won't be found in other flight stages.

At TO/L velocities are low —> constraints needed to prevent stall
↳ longitudinal aspects only here

- **Performance:** Aircrafts need to take-off from/land onto existing runways

↳ measure : minimum runway length L_r

Parameters involved in determining runway length necessary for a given aircraft:

- Aircraft velocity $\vec{V}$ (aero) and $\vec{V_k}$ (with respect to the ground : $\vec{V_k} = \vec{V} + \vec{V_w}$)
- Capability of the aircraft to accelerate/decelerate: $\Gamma \equiv$ acceleration (+) or deceleration (-)
- Slope of air phase: γ (small air phase including in runway length calculation)

What does the pilot do during take-off and landing:

- Stick, δ_m: pilot controls velocity : both at take off / landing

  For both take off and landing, the velocities **will be imposed**: the pilot needs to act to match them.

- Throttle, δ_x: pilot controls slope
  - **Landing.** Aircraft needs to touch down at the beginning of the runway.

↳ trajectory important —> strict objective
↳ pilot "plays" with δ_x
} more complicated, margins to stall will be higher

  - **Take-off.** Trajectory, slope objective less important: just need positive slope!

↳ goes full throttle and does not touch δ_x anymore

2 Take-off and landing phases and corresponding regulations

2.1 Take-off phase

A successful take off is comprised of three distinct phases, which will determine the runway length necessary for a given aircraft.

- **Acceleration:** Brings the aircraft from a zero velocity to the velocity required to lift wheels from the ground.

  ⇒ GROUND PHASE

- **Transition:** Rotation during which aircraft leaves the ground and places itself on a rectilinear climbing trajectory.

  ⇒ MIXED PHASE (GROUND + AIR)
  ↳ h_s = 35ft = 10.7 m

- **Rectilinear climb:** Lasts until a safety height is reached. AIR PHASE
  - Additional constraint: must reach h_s with $\gamma > \gamma_{min}$
  - During this phase, the landing gears are OUT

Note: After the safety height is reached, pilots retracts the landing gears, ≈ 3 s after leaving the ground
after that, initial climb starts –

SUCCESSFUL TO ≡ LEFT THE GROUND
(≠ REJECTED / ABORTED TAKE OFF)

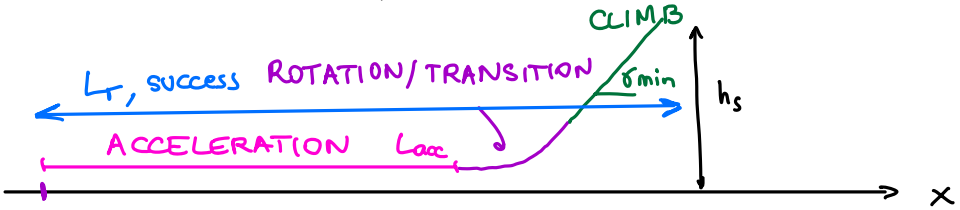
Take-off should be successful *even with one engine experiencing problems.* If the engine issue occurs **early enough**, take-off is interrupted. In this case, only two phases can be identified:

- **Acceleration:** till engine issue
- **Deceleration:** after engine issue

} both are ON the ground

Minimum runway length. We call L_r the minimum runway length necessary for take-off. L_r is estimated from the **3** phases of a successful take-off, AND/OR the **2** phases of an interrupted take-off.

SUCCESSFUL



ABORTED



In an interrupted take-off, we will call V_1 the **maximum velocity at which the problem on an engine can occur.** V_1 has a very large influence on the size of the runway:

- if engine failure occurs BEFORE V_1, aircraft can stop on the ground
- if engine failure occurs AFTER V_1, aircraft MUST leave the ground.

Parameters impacting L_r:

acceleration Γ Liftoff - V_{LOF} slope of air phase γ

Small L_r high Γ small V_{LOF} large γ

acceleration phase short air phase short

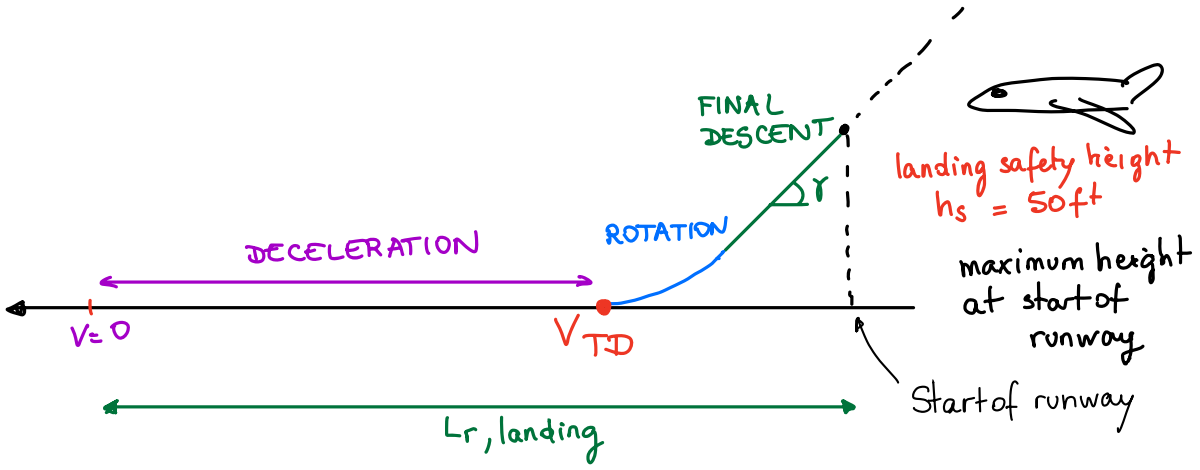
In the remainder of this chapter, we will establish relationships between those parameters and the actual value of L_r.



Figure 1: Boeing 747 during its take-off transition phase. The nose wheel is off the ground, but not the main wheels. The aircraft takes its take-off angle of attack, it is just before V_{LOF}.

2.2 Landing phase

(Successful!!) landing includes the same phases as a successful take-off, but in reverse order. The same parameters will therefore influence L_r at the landing stage.



2.3 Regulations for take-off and landing

Regulations (that have to be satisfied for an aircraft to be certified and sold) are mostly about velocities, slopes, and length of runway. We'll consider FAA regulations (FAR 25 for civilian aircrafts, FAR 23 for light aircrafts, military has its own regulations).

aerodynamic velocities → what the pilot sees

Successful take-off (includes "air" phase)

2 "MASTER" velocities

- **Min Velocity for longitudinal flight** → **Stall Velocity**
  - without ground effect → V_S → Easy to model → mostly depends on mass, high-lift systems, altitude
  - with ground effect → V_{mu} (minimum unstick) → Velocity below which it is not possible to get the aircraft of the ground → difficult to model → $V_{mu} \approx 0.95 - 0.99\, V_S$ (engine failure / engines ok)
- **Min Velocity for lateral flight**
  - on the ground → V_{mc_G} → Min velocity to maintain aircraft on runway axis in case of engine failure with rudder alone
  - in the air → V_{mc_A} → Min velocity to maintain $\psi = cst$ with less than 5° bank angle (ϕ) with one engine failure.

3 "OPERATIONAL" velocities (associated with pilot action)

- V_1 decision velocity → during acc. phase. if engin failure occurs before V_1, aircraft stops. else, must take off. → determined to minimize runway length → V_2 → Velocity at h_S with engine failure → $\begin{cases} V_2 > 1.2\, V_S \\ V_2 > 1.1\, V_{mc_A} \end{cases}$
- V_R rotation velocity → velocity at which pilots pulls on stick to bring aircraft to $\alpha_{take\text{-}off}$ → $\begin{cases} V_R > 1.1\, V_{mu} \\ V_R > 1.05\, V_{mc_A} \\ V_R > V_1 \end{cases}$ ← would be awkward to stop after rotation begins
- V_{LOF} → Velocity at which aircraft leaves the ground → $\begin{cases} V_{LOF} > 1.1\, V_{mu} \text{ (no e.g.)} \\ V_{LOF} > 1.05\, V_{mu} \text{ (with e.g.)} \end{cases}$

Interrupted (rejected) take-off

1 master velocity: V_{mc_G}

1 operational velocity: V_1

$\{ V_1 > V_{mc_G}$

Landing

1 master velocity: V_S

2 operational velocities

- V_{APP} Approach velocity = aircraft velocity at h_S → $V_{APP} > 1.3\, V_S$
- V_{TD} touch-down velocity → $V_{TD} > 1.15\, V_S$

Slopes Take-off is successful if aircraft reaches the safety height with a positive slope, even with engine failure. *Minimal slopes required at take-off.*

$\gamma_{\min}$	Landing gears out	Landing gears retracted
twin-engine	0%	2.4%
3 engines	0.3%	2.7%
4 engines	0.5%	3%

For landing, there is no regulation on the slope, but the standard ILS is 3 degrees (slope will be different depending on airport)

Runway length Regulations impose the length of the runway to be longer than the max of the following three lengths:

- Distance to go through acceleration/deceleration and stop with engine failure;
- Runway length necessary to take off with engine failure; or
- 1.15 times the runway length necessary to take off without engine failure.

V_1 is typically estimated by assuming the first two lengths are equal, in which case the length of the runway is the distance needed to stop when engine failure occurs at V_1.

Runway length for landing is estimated for dry and hard surfaces, divided by 0.6 to account for harsher conditions or lack of precision on touchdown location. Landing distances are typically shorter than take-off.

3 Parameters influencing velocities and acceleration

Both velocity and acceleration have a first-order effect on runway length.

3.1 Kinematic velocity V_k

- Velocity with respect to the ground ($V_{k,\text{LOF}}$, $V_{k,TD}$) at take-off and landing.
- V_k small $\Rightarrow$ L_r small
- $\overrightarrow{V}_k = \overrightarrow{V} + \overrightarrow{V}_w$
  - Neglecting the aircraft's angle of attack during acceleration, and denoting by γ_r the slope of the runway, the aerodynamic velocity is defined as (**S**):

Anything that reduces V_{LOF}, that is, the aerodynamic velocity at which aircraft can leave the ground, will decrease L_r

$$V^2 \approx \frac{mg}{S}\frac{\cos\gamma_r}{\frac{1}{2}\rho C_z}\left(-\underbrace{\frac{\frac{F}{S}\sin\alpha_r}{\frac{1}{2}\rho C_z}}_{\text{smaller}}\right) \tag{1}$$

Effect of the environment

- **Wind:** With a head wind, $V_k = V - |V_w|$, so the magnitude of V_k decreases for a given aerodynamic velocity. beneficial for L_r
- **Altitude:**

Low altitude $\rightarrow$ higher ρ $\rightarrow$ lower V $\rightarrow$ lower L_r

- **Runway slope:** little influence on aerodynamic velocity, but a clear effect on the aircraft ability to accelerate or decelerate (see below)

Effect of the aircraft

- **Wing loading:** $\frac{W}{S} = \frac{mg}{S}$ low wing loading ⇒ short runway
  - gliders/ early designs (1900) $\frac{W}{S} \simeq 1000\ N/m^2$
  - typical transport aircrafts $\frac{W}{S} \approx 6000\ N/m^2$ (high value comes from velocity optimization for cruise)

- **High-lift systems:**

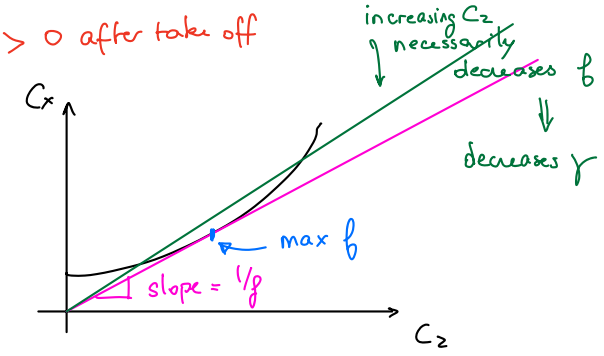
flaps/slats increase $C_{z_{max}}$ ⇒ can decrease V_{LOF}

needs to be > 0 after take off

BUT $F - D = mg\gamma$ (P)

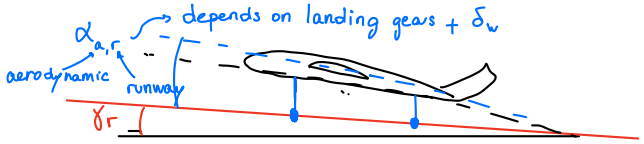
$L = W \Rightarrow F - \frac{mg}{f} = mg\gamma$

lift to drag ratio $\frac{C_z}{C_x}$



Configuration	$C_{z_{TO}}$	C_{x_0}	f	$fC_{z_{TO}}$	$C_{z_{\max}}$
Nothing deployed	0.97	0.02	15.4	14.9	1.4
Slats only, 17°	1.2	0.029	12.7	15.2	1.73
Slats 17°, Flaps 15°	1.55	0.036	10.7	16.6	2.23
Slats 25°, Flaps 22°	1.85	0.042	9.4	17.4	2.66
Slats 25°, Flaps 41°	2.16	0.076	7.5	16.2	3.11

Table 1: Typical values for take-off (extra drag from landing gears not accounted for).



3.2 Acceleration

Acceleration or deceleration is expressed as:

$\underbrace{m\Gamma}_{\parallel \text{runway}} = \text{Thrust} - \text{Drag} - \text{Friction} - \text{Weight projected onto runway}$

$F\cos\alpha_{a,r} \quad \frac{1}{2}\rho S V^2 C_x \quad f_r R$

R: normal reaction force → lift created; weight of plane; thrust projected ⊥ runway

friction coefficient: σ_r when accelerating; μ_r when decelerating

The final expression for the acceleration can be re-written:

$$\Gamma = \frac{F}{m}\left(\cos\alpha_{a,\mathrm{r}} + f_r \sin\alpha_{a,\mathrm{r}}\right) - \frac{\frac{1}{2}\rho S V^2}{m}\left(C_x - f_r C_z\right) - g\left(\sin\gamma_r + f_r\cos\gamma_r\right) \tag{2}$$

4 Runway length calculation - General considerations

For both take off and landing, the aircraft needs to go (one way or another) from a zero velocity to a standard rotation velocity V_R or touch down V_{TD} (we'll call this velocity V_{ref}). We need to integrate between those two velocity a differential equation for the distance, which is obtained from the definition of the acceleration:

$$\Gamma = \frac{dV}{dt} = \frac{dV}{dx}\frac{dx}{dt} = V\frac{dV}{dx} = \frac{1}{2}\frac{dV^2}{dx} \tag{3}$$

similar

V_{LOF} (or V_R) for take off or V_{TD} for landing

leading to

$$dx = \frac{1}{2}\frac{d(V^2)}{\Gamma}$$

$$L_r^{acc} = \int_0^{L_r^{acc}} dx = \frac{1}{2}\int_0^{V_{REF}} \frac{dV^2}{\Gamma}$$

- Take-off: Γ is positive, integration from 0 to take-off velocity (V_{LOF} or V_R)
- Landing: Γ is negative, but integration bounds are reversed: L_r stays positive.

The expression for Γ is complicated. There may not even be an analytical expression! Instead of the actual expression, a constant, "average" acceleration is often used:

Best "average acceleration" $\Gamma_{0.7\,V_{ref}}$ ≡ acceleration evaluated when $V = 0.7\,V_{REF}$

$$L_r^{acc} = \frac{1}{2}\frac{V_{REF}^2}{\Gamma_{0.7\,V_{REF}}}$$

to/from h_s

total runway length : $L_r^{TO} = L_r^{acc} + L_r^{flare} + L_r^{climb}$

$L_r^{L} = L_r^{dec} + L_r^{flare} + L_r^{descent}$

5 Runway length calculation - Take-off

5.1 Acceleration phase

This phase occurs at the maximum allowed thrust, in which case we typically have a ratio thrust over weight ≈ 0.3 for a transport aircraft, up to 1 for a military aircraft. To maximize acceleration, it is possible to optimize the aircraft's angle of attack for this phase, $\alpha_{a,r}$. It is useful to first look at the friction coefficient, σ_r in this case (f_r in general).

Friction coefficient during acceleration, σ_r

At take-off, wheels are not used for accelerating not braking. But the uneven pressure distribution inside the tire (higher at the front) creates a force that resists motion. This force T is proportional to the ground normal reaction:

$$T = \sigma_r R_n \tag{4}$$



The proportionality constant is the friction coefficient σ_r, which increases with velocity. It is not a constant:

$$\sigma_r = \sigma_{r,0} + a_r \mathcal{L} V^2 \tag{5}$$

$\sigma_{r,0}$: friction coefficient at zero velocity

$\mathcal{L}$: load = weight - lift $= mg\cos\gamma_r - \frac{1}{2}\rho S V^2 C_z$

a_r: constant, empirical function of { # of wheels n_r, diameter of wheels D_r, tire pressure p_r }

e.g. $a_r = \dfrac{4\times10^{-4}}{g\, n_r\, p_r\, D_r}$

0.014 rough, dry cement

0.053 grass

Boeing 747 at take-off velocity

at 70 m/s → 250 km/h : $\sigma_r = 1.68\,\sigma_{r_0}$

Optimal angle of attack during acceleration, $\alpha_{a,r}$

The term $A = C_x - \sigma_r C_z$ is the aerodynamic contribution to the acceleration Γ. It will be minimum for:

(⊖ sign in Γ expression : we want to maximize Γ)

$$\frac{dA}{dC_z} = 0 = \frac{dC_x}{dC_z} - \sigma_r$$

$$\Rightarrow \frac{dC_x}{dC_z} = \sigma_r$$

Drag polar : $C_x = C_{x_0} + k C_z^2$, with $k \approx \dfrac{1}{\pi AR}$

$C_z \approx \dfrac{2\pi}{1+\frac{2}{AR}}\,\alpha \approx \alpha_{a,r}$ (neglecting α_0)

$$\frac{dC_x}{dC_z} = 2k C_z = \frac{4}{AR+2}\,\alpha_{a,r} = \sigma_r$$

$\alpha_{a,r}$: angle of attack minimizing aero term in acceleration

$$\alpha_{ar} = \frac{AR+2}{4}\,\sigma_r$$

for $AR = 10$, $\alpha_{a,r} = 3\sigma_r$

* cement : $\alpha_{a,r} \approx 3 \times 0.02 = 0.06$ rad $= 3.4°$
* meadow : $\alpha_{a,r} \approx 3 \times 0.05 = 0.15$ rad $= 8.6°$ ⇐ HIGH!

Aero term important at high velocities, when aircraft close to take off.

On the other hand, $\alpha_{a,r}$ contributes to thrust too, we need to maximize:

$$B = \cos\alpha_{a,\mathrm{r}} + \sigma_r \sin\alpha_{a,\mathrm{r}} \tag{6}$$

$\frac{dB}{d\alpha_{a,r}} = 0 = -\sin\alpha_{ar} + \sigma_r \cos\alpha_{ar}$

$\Rightarrow \tan\alpha_{ar} \approx \alpha_{a,r} = \sigma_r$

↑ angle of attack maximizing thrust contribution to acceleration, weakly dependent on velocity

Clearly, we need to find a compromise between the two contributions! With some algebra, for an aircraft such as the one used in the case studies, we find that the optimal value for $\alpha_{a,r}$ is the average between the two:

$\alpha_{a,r}^{\mathrm{OPT\ (compromise)}} \approx \frac{3\sigma_r + \sigma_r}{2} = \boxed{2\sigma_r}$

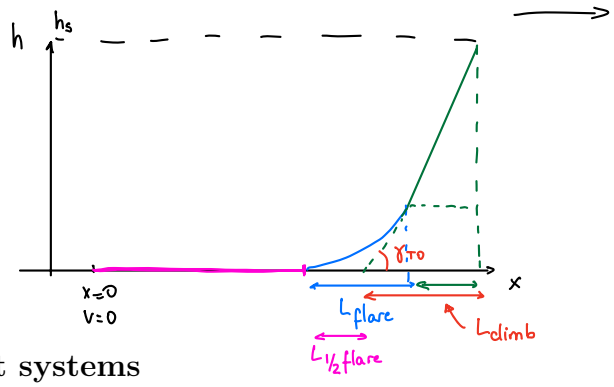
may need to be changed based on aircraft geometry

5.2 Flare

What we'll call take-off flare has two phases: one is on the ground, and starts at V_R when the pilot pulls on the stick to bring the aircraft from $\alpha_{a,r}$ to the take-off angle of attack $\alpha_{a,\mathrm{LOF}}$. This phase stops when the aircraft reaches V_{LOF} and leaves the ground. The second phase then starts, and the aircraft effectively follows a curved trajectory until it reaches the appropriate slope for straight climb. This second phase stops when the aircraft starts its straight climb. With some assumptions (for example, constant rotation velocity and acceleration), we can show that the length of this phase is

$$L_{1/2\ \mathrm{flare}} = \frac{1}{n_f - 1}\frac{V_{\mathrm{LOF}}^2}{2g}\gamma_{TO} \tag{7}$$

where n_f is the load factor (typical value: 1.1), γ_{TO} is the slope of the initial climbing phase. The duration of this phase is around 3 seconds.



5.3 Climb slope - high-lift systems

To optimize the climb slope γ, for a given thrust, the lift-to-drag ratio must be maximized (ratio of **P** and **S**):

$$\frac{F}{mg} = \frac{1}{f} + \gamma \tag{8}$$

f: lift to drag ratio

Regulations impose a minimum slope at h_s, in the case of engine failure. Therefore, if F is the thrust provided by all engines, an aircraft with n engines must satisfy:

total thrust that can be provided by all engines

$$\frac{n-1}{n}\frac{F}{mg} = \frac{1}{\left(\frac{L}{D}\right)_{\text{initial climb}}} + \gamma_{min}$$

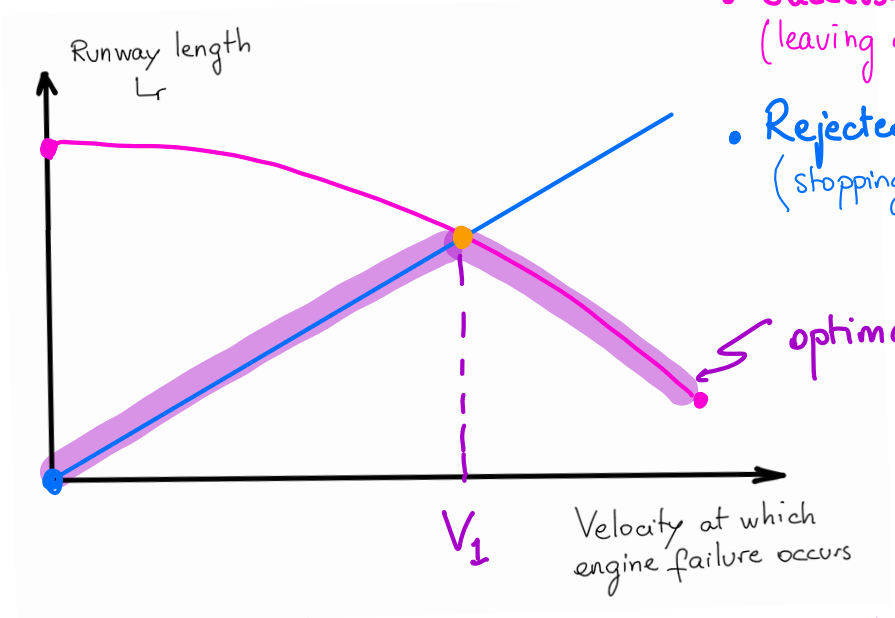
γ_{min}: minimal allowed slope

n: # of engines

in practice, want a ">" instead of "="

5.4 Interrupted (Reject) Take-off RTO

If engine failure occurs at low velocity, it is better to stop the aircraft on the runway. If the velocity is too high, the aircraft will have to take off with its faulty engine. This is sketched in the following L_r vs. Velocity plot:



- Successful take-off (leaving ground no matter what)
- Rejected take-off (stopping on the ground no matter what)

optimal runway length

largest L_r occurs when engine fails AT V1

Reject take-off typically done with spoilers (kill lift and increase drag very quickly)

↳ V_1 allows for pilot reaction time (≈ 1 sec)

$V_1 = V_{\text{effective engine failure}} + \underbrace{\Delta V}_{\text{reaction time}}$

5.5 Take-off runway length

The total runway length is obtained by adding all contributions together. To get a faster estimate, an semi-empirical formula exists, called the "magic line":

$$L_r = (0.23 - n\ 0.01)\ \tilde{L} + \boxed{(244 - 42n)} \quad [\text{m}]$$

(there exists more complicated expressions)

with

* $\tilde{L} = \dfrac{(mg)^2}{\sigma^{0.8}\, S C_{z_{max}} F_{0.7}}$

* $n \equiv$ number of engines
* $\sigma \equiv$ ratio of densities ρ/ρ_0 ← standard sea level
* $F_{0.7} \equiv$ thrust delivered by all engines at $V = 0.7\, V_{LOF}$

⇒ Actual runway length when engine failure occurs at V_1
FAR 25 length: includes aerial phase for successful take-off.

□ constant term comes from near zero acceleration during climb in case of engine failure

↳ modification WITHOUT ENGINE FAILURE

$L_r = 0.196\ \tilde{L} + 31$ (includes a 15% safety margin)

actual length:

$$L_{r,\text{actual}} = \frac{L_r}{1.15}$$

6 Runway length calculation - Landing

6.1 Descent slope

High-lift systems

During landing, the slope is negative: compared to take-off, for the same lift-to-drag ratio, the required thrust will be much lower. In contrast to the take-off situation, it is therefore possible to "afford" the sharp decrease in lift-to-drag ratio when high-lift systems are fully deployed, with the benefit of decreasing as much

as possible the approach velocity (Remember that at take-off, climbing may not be possible when high-lift systems are deployed too much, as there would not be enough thrust in **P** to get a positive climb slope).

The high-lift systems are sometimes complemented by the deployment of spoilers, further decreasing the lift-to-drag ratio. This may prove useful in case the aircraft needs to abort landing and immediately climb back again: the spoilers can be retracted nearly instantaneously, which would immediately decrease the drag and increase the thrust balance.

Optimal descent/approach slope

To minimize the runway length, given the fact that it starts when the aircraft passes the safety height h_s, it is useful to consider the slope in approach. This has two competing effects: the more negative the slope,1) the shorter the ground projection of the linear descent will be, but 2) the longer the landing flare will be too.

In order to evaluate the optimal descent slope, let's consider a simple model for the runway length needed till touch-down:

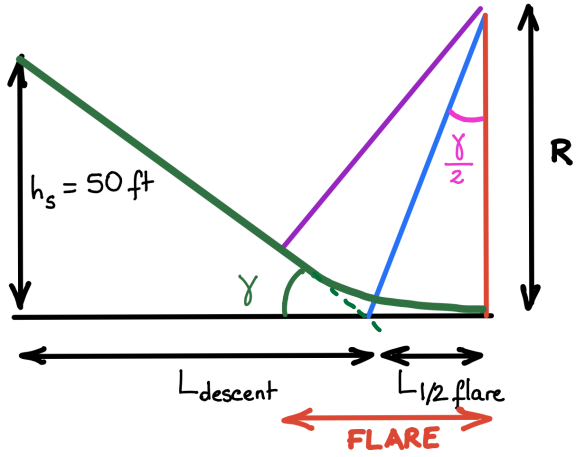
$$L_{r,\mathrm{TD}} = L_{\mathrm{descent}} + L_{1/2\ \mathrm{flare}} \tag{9}$$

We have

$$\tan\gamma = \frac{h_s}{L_{\mathrm{descent}}} \tag{10}$$

$$L_{1/2\ \mathrm{flare}} = R\frac{\gamma}{2} \tag{11}$$

where R is the curvature radius of the flare (see schematic). We will assume for simplicity that the velocity is constant throughout the flare.



The trajectory gets curved by the introduction of a force perpendicular to the velocity. This force is provided by the increase in lift ΔC_z created by pulling on the stick. It can easily be shown by introducing just a bit of dynamics (see later chapters) that the curvature radius can be written

$$R = \left(\frac{\Delta C_z}{C_z}\right)^{-1}\frac{V^2}{g} = \frac{1}{n_f - 1}\frac{V^2}{g} \tag{12}$$

leading to

$$L_{1/2\ \mathrm{flare}} = R\frac{\gamma}{2} = \frac{1}{n_f - 1}\frac{V^2}{g}\frac{\gamma}{2} \approx 5\frac{V^2}{g}\gamma \tag{13}$$

and

$$L_{r,\mathrm{TD}} = \frac{h_s}{\gamma} + \frac{1}{n_f - 1}\frac{V^2}{g}\frac{\gamma}{2} \tag{14}$$

As usual, the optimal slope will be obtained by differentiating $L_{r,\mathrm{TD}}$ wrt γ and setting the resulting expression to 0, leading to:

$$\gamma_{\mathrm{opt}} = \sqrt{2h_s g(n_f - 1)}\frac{1}{V} \approx \frac{5.46}{V} \tag{15}$$

where $h_s = 50$ ft $= 15.2$ m. For an approach velocity of 120 kt, the optimal slope obtained with this calculation is therefore 0.09 rad or 5.2°. It would decrease to 4.4°at 140 kt, not far from the standard 3°of ILS. This calculation does not fully account for the dynamics of the system, but provides the right order of magnitude.

Note that the more negative the slope, the higher the start of the flare, and the larger the uncertainty on the exact location of the touchdown. For a typical landing with a classic autopilot, the horizontal dispersion on TD is ±320 m in 99.7% of all landings, down to ±100 m for 67% of all landings.

6.2 Friction coefficient during deceleration, μ_r

After touch-down, the aircraft must decelerate efficiently, mostly thanks to its brakes. In this context, the braking coefficient μ_r, which relates the (tangential) braking force and the aircraft load $\mathcal{L}$ (weight - lift) will play a predominant role:

$$T = \mu_r \mathcal{L} \tag{16}$$

We'll assume that μ_r can be expressed as a function of the velocity:

$$\mu_r = \frac{\mu_{r,\max}}{1 + \frac{V}{W_r}} \tag{17}$$

where typical values for the parameters $\mu_{r,\max}$ and W_r are provided in the table below:

Runway	$\mu_{r,\max}$	W_r
Dry, clean cement	0.9	100 m/s
Wet, clean cement	0.7	50 m/s
Iced	0.15	25 m/s

Note that $\mu_{\max}$ is obtained when braking involves the right amount of skidding, controlled via an anti-lock braking system (ABS).

6.3 Deceleration

Recall the expression for the deceleration Γ:

$$\Gamma = \frac{F}{m}\left(\cos\alpha_{a,\mathrm{r}} + \mu_r \sin\alpha_{a,\mathrm{r}}\right) - \frac{\frac{1}{2}\rho S V_a^2}{m}\left(C_x - \mu_r C_z\right) - g\left(\sin\gamma_r + \mu_r \cos\gamma_r\right) \tag{18}$$

To increase braking efficiency, it is beneficial to destroy lift right after touch-down, so that the load (weight - lift, $\mathcal{L}$) is maximum. This is done by deploying spoilers, that decrease C_z and increase C_x. Typical decelerations are around $0.5g$, and typical maximum around $0.9g$. The main contribution is $g\mu_r \cos\gamma_r \approx g\mu_r$: reducing Eq. 18 to this term allows to obtain quick estimate of the length needed to stop.

6.4 Runway length at landing

The length can be calculated by integrating the motion equation. A faster approach similar to the magic line is provided here:

$$L_r = 0.5V_S^2 + 223 = \frac{L_{r,\mathrm{actual}}}{0.6}\,, \tag{19}$$

where a safety factor of 1/0.6 is built in. It assumes that the aircraft has ABS and automatic deployment of the spoilers (relying on pilot action can add ≈ 2 seconds delay, or ≈ 120m for the aircraft used in the case study).